

COMPARATIVE ANALYSIS

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1. Object Detection Performance Comparison

Two object detection models, Viola–Jones and YOLO, produce the following results: Model A has 80 True Positives, 20 False Positives, and 30 False Negatives, while Model B has 90 True Positives, 15 False Positives, and 20 False Negatives.

- (a) Calculate the precision and recall for both models.
- (b) Compare the results and determine which model provides better overall object detection performance. Justify your answer based on the calculated values.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce the following displacement errors in pixels: Method A: 2, 3, 2 and Method B: 3, 5, 4 for three consecutive frames.

- (a) Calculate the average displacement error for both methods.
- (b) Compare the two methods and determine which method provides more accurate motion estimation.
- (c) Justify your answer based on the error values.

3. Depth Estimation Comparison

A stereo vision system and a Structure-from-Motion (SfM) system estimate the depth of three objects. The estimated depths are: Stereo Vision: 2.0 m, 3.0 m, 4.0 m and SfM: 2.5 m, 3.8 m, 5.0 m, while the actual depths are 2 m, 3 m, and 4 m, respectively.

- (a) Calculate the absolute error for each object for both methods. (b) Calculate the average absolute error for both methods.
- (c) Compare the methods and identify which provides more accurate depth estimation.

1. Object Detection Performance Comparison

Given Data

Model A (Viola-Jones):

True Positives (TP) = 80

False Positives (FP) = 20

False Negatives (FN) = 30

Model B (YOLO):

True Positives (TP) = 90

False Positives (FP) = 15

False Negatives (FN) = 20

(a) Calculate Precision and Recall

Precision Formula:

Precision = $TP / (TP + FP)$

Recall Formula:

Recall = $TP / (TP + FN)$

Model A – Viola-Jones

Precision = $80 / (80 + 20)$

= $80 / 100$

= 0.80

Precision = 80%

Recall = $80 / (80 + 30)$

= $80 / 110$

= 0.7273

Recall = 72.73%

Model B – YOLO

Precision = $90 / (90 + 15)$

= $90 / 105$

= 0.8571

Precision = 85.71%

Recall = $90 / (90 + 20)$

= $90 / 110$

= 0.8182

Recall = 81.82%

(b) Comparison

Model A (Viola-Jones):

Precision=80%

Recall = 72.73%

Model B (YOLO):

Precision=85.71%

Recall = 81.82%

(c) Justification

YOLO provides better overall object detection performance because it has higher precision and higher recall than Viola-Jones.

YOLO Precision = 85.71% > 80%

YOLO Recall = 81.82% > 72.73%

YOLO also has more True Positives (90) and fewer False Positives (15) and False Negatives (20).

Therefore, YOLO provides better overall object detection performance.

2. Motion Estimation Error Analysis

Given Data

Method A displacement errors = 2, 3, 2 pixels

Method B displacement errors = 3, 5, 4 pixels

(a) Calculate Average Displacement Error

Method A:

$$\text{Average Error} = (2 + 3 + 2) / 3$$

$$= 7 / 3$$

$$= 2.33 \text{ pixels}$$

$$\text{Average Error of Method A} = 2.33 \text{ pixels}$$

Method B:

$$\text{Average Error} = (3 + 5 + 4) / 3$$

$$= 12 / 3$$

$$= 4 \text{ pixels}$$

$$\text{Average Error of Method B} = 4 \text{ pixels}$$

(b) Comparison

Method A average error = 2.33 pixels

Method B average error = 4 pixels

Since a smaller displacement error indicates more accurate motion estimation, Method A is more accurate.

(c) Justification

Method A has an average displacement error of 2.33 pixels, while Method B has an average error of 4 pixels.

Difference in average error:

$$4 - 2.33 = 1.67 \text{ pixels}$$

Therefore, Method A has lower error and provides more accurate motion estimation.

Final Answer: Method A provides more accurate motion estimation.

3. Depth Estimation Comparison

Given Data

Actual depths = 2 m, 3 m, 4 m

Stereo Vision estimates = 2.0 m, 3.0 m, 4.0 m

SfM estimates = 2.5 m, 3.8 m, 5.0 m

(a) Calculate Absolute Error

Formula:

$$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$$

Object 1

Stereo Vision:

$$|2.0 - 2.0| = 0 \text{ m}$$

SfM:

$$|2.5 - 2.0| = 0.5 \text{ m}$$

Object 2**Stereo Vision:**

$$|3.0 - 3.0| = 0 \text{ m}$$

SfM:

$$|3.8 - 3.0| = 0.8 \text{ m}$$

Object 3**Stereo Vision:**

$$|4.0 - 4.0| = 0 \text{ m}$$

SfM:

$$|5.0 - 4.0| = 1.0 \text{ m}$$

Absolute Error Table

Object 1:

Stereo

Vision=0m

$$\text{SfM} = 0.5 \text{ m}$$

Object 2:

Stereo

Vision=0m

$$\text{SfM} = 0.8 \text{ m}$$

Object 3:

Stereo

Vision=0m

$$\text{SfM} = 1.0 \text{ m}$$

(b) Calculate Average Absolute Error

Stereo Vision:

$$\text{Average Error} = (0 + 0 + 0) / 3$$

$$= 0 / 3$$

$$= 0 \text{ m}$$

Average Absolute Error of Stereo Vision = 0 m

SfM:

$$\text{Average Error} = (0.5 + 0.8 + 1.0) / 3$$

$$= 2.3 / 3$$

$$= 0.7667 \text{ m}$$

Average Absolute Error of SfM = 0.77 m

(c) Comparison and Conclusion

Stereo Vision average absolute error = 0 m

SfM average absolute error = 0.77 m

Since lower error indicates better depth estimation accuracy, Stereo Vision is more accurate.

Stereo Vision produces exactly the actual depth values for all three objects, resulting in zero error.

Therefore, Stereo Vision provides more accurate depth estimation than SfM.